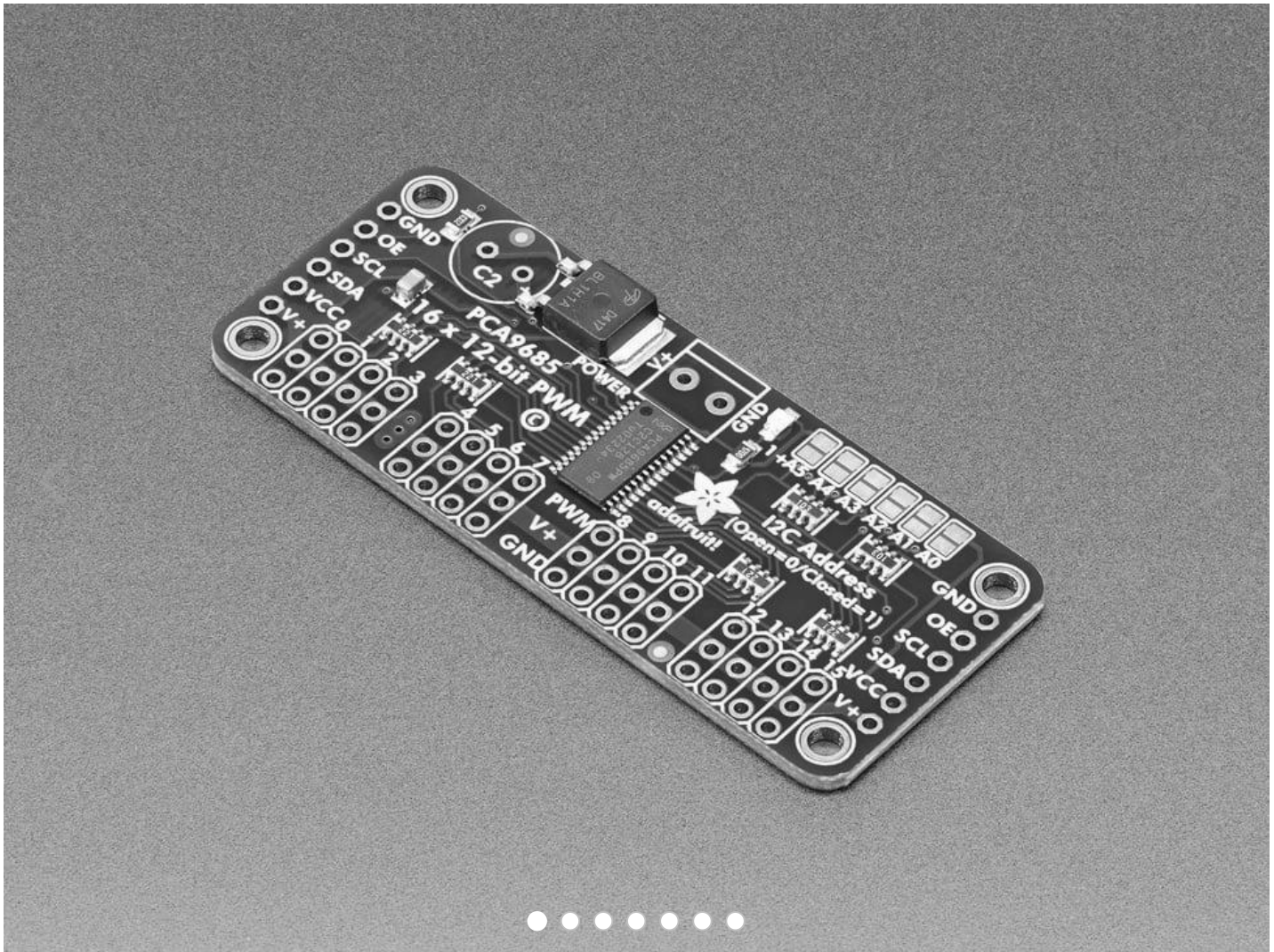


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Robotics & CNC / Servos

Adafruit 16-Channel 12-bit PWM/Servo Driver - I2C interface - PCA9685

Product ID: 815

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1

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Description

You want to make a cool robot, maybe a hexapod walker, or maybe just a piece of art with a lot of moving parts. Or maybe you want to drive a lot of LEDs with precise PWM output. Then you realize that your microcontroller has a limited number of PWM outputs! What now? You could give up OR you could just get this handy PWM and Servo driver breakout.

When we saw this chip, we quickly realized what an excellent add-on this would be. **Using only two pins, control 16 free-running PWM outputs!** You can even chain up 62 breakouts to control up to 992 PWM outputs (which we would really like to see since it would be glorious)

- It's an I2C-controlled PWM driver with a built-in clock. That means that, unlike the TLC5940 family, you do not need to continuously send it signal tying up your microcontroller, it's completely free running!
- It is 5V compliant, which means you can control it from a 3.3V microcontroller, with 3.3V logic pull-ups, and still safely sink up to 5.5V outputs (this is good for when you want to control common-anode white or blue LEDs with 3.4+ forward voltages)
- 6 address select pins so you can wire up to 62 of these on a single i2c bus, a total of 992 outputs - that's a lot of servos or LEDs
- Adjustable frequency PWM up to about 1.6 KHz
- 12-bit resolution for each output - for servos, that means about 4us resolution at 60Hz update rate
- Configurable push-pull or open-drain output
- Output enable pin to quickly disable all the outputs

We wrapped up this lovely chip into a breakout board with a couple of nice extras

- Terminal block for power input (or you can use the 0.1" breakouts on the side)
- Reverse polarity protection on the terminal block input. The terminal block included with your product may be blue or black.
- Green power-good LED
- 3 pin connectors in groups of 4 so you can plug in 16 servos at once (Servo plugs are *slightly* wider than 0.1" so you can only stack 4 next to each other on 0.1" header)

- "Chain-able" design
- A spot to place a big capacitor on the V+ line (in case you need it)
- 220 ohm series resistors on all the output lines to protect them, and to make driving LEDs trivial
- Solder jumpers for the 6 address select pins

This product comes with a fully tested and assembled breakout as well as 4 pieces of 3x4 male straight header (for servo/LED plugs), a 2-pin terminal block (for power), and a piece of 6-pin 0.1" header (to plug into a breadboard). A little light soldering will be required to assemble and customize the board by attaching the desired headers but it is a 15-minute task that even a beginner can do. If you want to use right-angle 3x4 headers, we also carry a 4 pack in the shop.

[Check out our tutorial with CircuitPython & Arduino libraries/examples, wiring diagrams, schematics, Fritzing, and more!](#)

Technical Details

Specifications:

- Dimensions (no headers or terminal block) 2.5" x 1" x 0.1" (62.5mm x 25.4mm x 3mm)
- Weight (no headers or terminal block): 5.5grams
- Weight (with 3x4 headers & terminal block): 9grams
- This board/chip uses I2C 7-bit address between 0x40-0x7F, selectable with jumpers

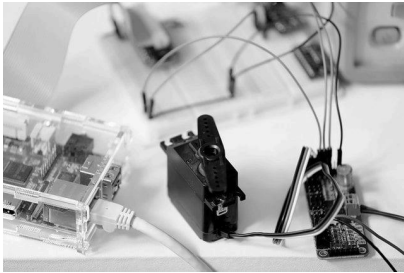
Datasheets, EagleCAD PCB files, Fritzing object and more in the tutorial!

Revision History:

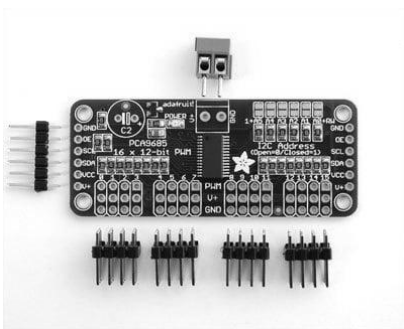
- **As of December 20, 2022** – we've updated this PCB with [Adafruit Penguin](#) to make a lovely and legible silkscreen - you may get the new PCB or the older version with vector fonts - both are identical other than the fancy silkscreen.
- As of Feb 16, 2015 we now use a much beefier polarity-protection transistor. Functionality is otherwise the same



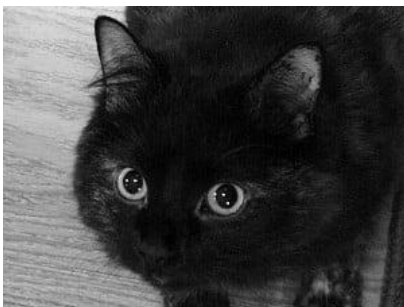
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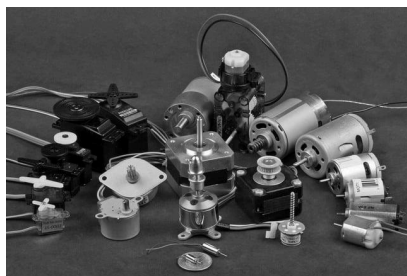
Adafruit 16 Channel Servo Driver with Raspberry Pi
Easily drive many servos in your projects



Adafruit PCA9685 16-Channel Servo Driver
Drive many servos easily



Raspberry Pi Wifi-Controlled Cat Laser Toy
Make your own telepresence cat toy, perfect for exercising/torturing your favorite feline

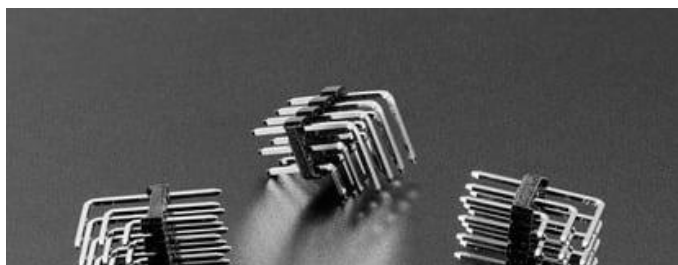


Adafruit Motor Selection Guide

Choose the right motor (and controller) for the job!

[See All Guides](#) 

May We Also Suggest...



3x4 Right Angle Male Header - 4 pack



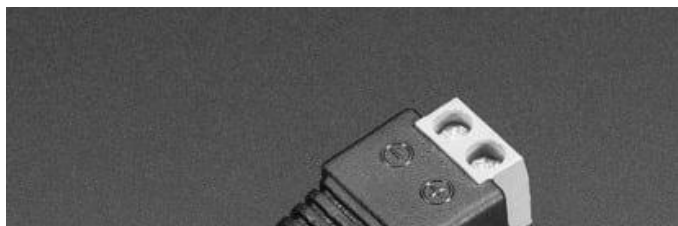
5V 10A switching power supply



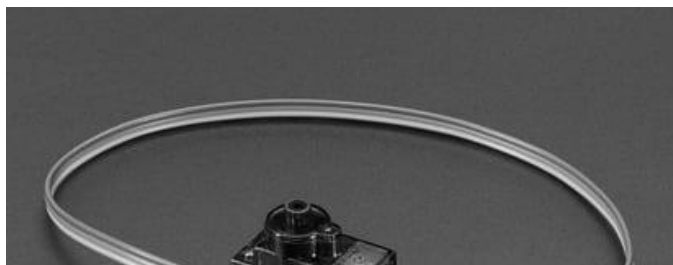
5V 2A (2000mA) switching power supply - UL Listed



Continuous Rotation Servo - FeeTech FS5103R



Female DC Power adapter - 2.1mm jack to screw terminal block



Micro servo - TowerPro SG92R



Standard servo - TowerPro SG-5010 - 5010



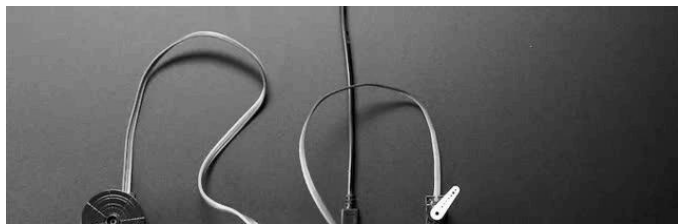
Adafruit ATtiny817 Breakout with seesaw - STEMMA QT / Qwiic



Adafruit 16-Channel 12-bit PWM/Servo Shield - I2C interface



Unassembled Mini Pan-Tilt Kit (Without Micro Servos)



Adafruit 16-Channel PWM / Servo HAT for Raspberry Pi - Mini Kit



Feedback 360 Degree - High Speed Continuous Rotation Servo

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